Calculations

Task 1: Campaign Viability Analysis

Brand Details:

Field	Value
Product	Premium Face Serum
Selling Price (SP)	₹800 per unit
Cost of Goods Sold (COGS)	₹280 per unit
Amazon Referral Fee	12% of SP
Monthly ad spend	₹3,00,000
Units sold via ads (monthly)	1,500
Organic units sold (monthly)	2,000
Fixed costs (monthly)	₹1,50,000 (warehousing, team, software)

Questions to Answer:

Q1. **Unit Economics:**

Gross Margin : Formula: Gross Margin = SP - COGS

Gross Margin => 800 - 280 = **520₹**

Amazon Fee: Formula: Amazon Fee = SP × Fee %

Amazon Fee => 800 * 12% = **96₹**

Ad cost per Unit: Formula: Ad Cost per Unit = Total Ad Spend ÷ Units Sold via Ads

Amazon Fee => 3,00,000 / 1500 = **200**₹

Net Margin per Unit for Ad-Attributed Sales: Formula: SP - - Product Cost - Amazon Fee - Ad Cost Per Unit

Net Margin => 800 - 280 - 96 - 200 = **224₹**

Q2. **Monthly Profitability:**

Formula: Monthly Profit = (Ad profit × Organic Profit) - Fixed Costs

Organic margin per unit:

800−280−96=424

Now:

424×2000=848000

So:

336000+848000−150000=**10,34,000₹**

Q3. **ACOS and TACOS:**

ACOS: Formula: ACOS = Ad Spend ÷ Ad Revenue × 100

ACOS => (3,00,000 / (1500*800))*100 = **25%**

TACOS: Formula: TACOS = Total Ad Spend ÷ Total Revenue (organic + paid) × 100

TACOS => (3,00,000/(1500+2000)*800)*100 = **10.71%**

Q4. **Break Even Ad-Spend:**

Total contribution BEFORE ad spend: Selling price - COGS - Amazon fee **(424 , previously calculated)**

Total Units: 1500+2000 = 3500

Maximum affordable ad spend: (424*3500)- 1,50,000 = 13,34,000

Break-even ACOS = (Max Affordable Ad Spend ÷ Ad Revenue) × 100

Break-even ACOS: (13,34,000 / 12,00,000)*100 = **111.1%**

Q5. **Scenario , Ad Efficiency Drop:**

ACOS: 40% , **Ad Units:** 1200 , **New Ad Revenue:** 1200*800 = 9,60,000

ACOS: Formula: ACOS = Ad Spend ÷ Ad Revenue × 100

40 * 9,60,000 = **Ad Spend => 3,84,00,000₹**

Ad Cost Per Unit: 3,84,000 ÷ 1200 = 320₹

Net Margin Per Ad Unit: 800 - 280 - 96 - 320 = 104₹

Total Ad Profit: 104 × 1200 = 1,24,800₹

Organic Profit: (800 - 280 - 96) × 2000 = 424 × 2000 = 8,48,000₹

Monthly Profit: 1,24,800 + 8,48,000 - 1,50,000 **= 8,22,800₹**

Q6. **Scenario , Scaling Up:**

The business remains profitable even after ACOS worsens to 40%, mainly because strong organic sales continue supporting overall profitability. However, ad efficiency declines significantly as advertising cost per unit increases from ₹200 to ₹320 while ad-attributed unit sales also decrease.

This suggests that the company should optimize campaigns, improve targeting, and reduce wasted ad spend before scaling further.